



Knowing When to Look Again

Dynamic risk of structural valve deterioration to personalise echo surveillance
after bioprosthetic aortic valve replacement

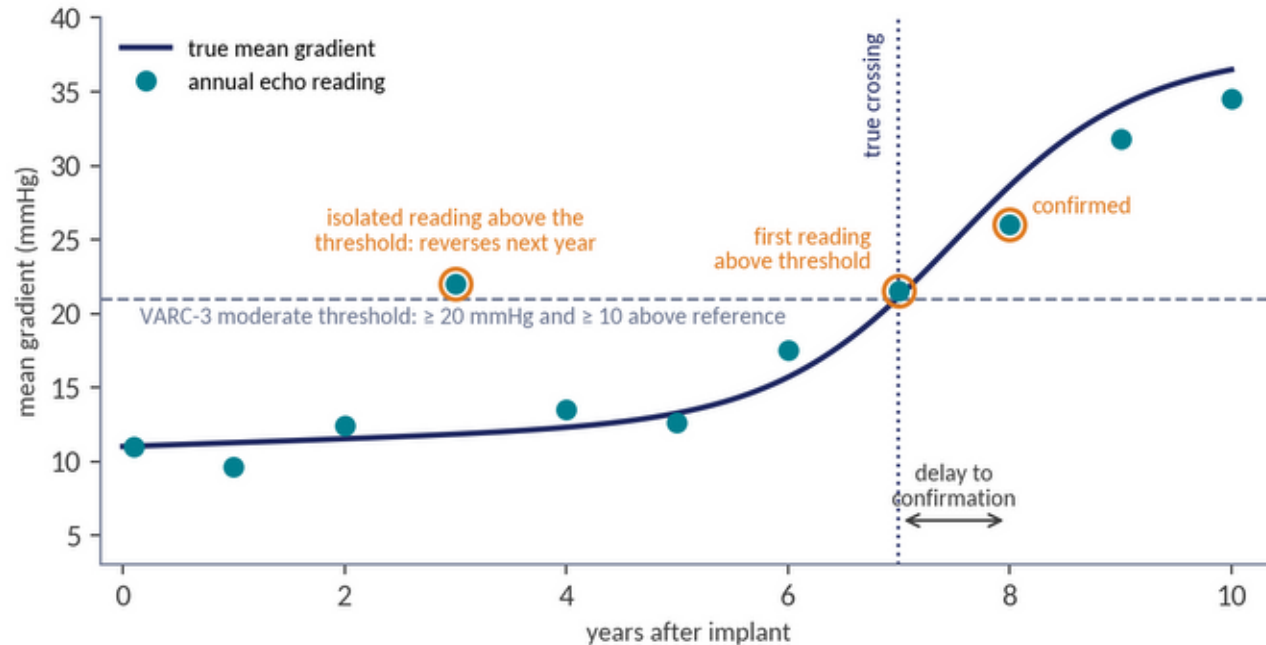
Team CardioNTUA

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The clinical problem

Deterioration is slow, silent, and hard to confirm



Schematic of one valve: the true gradient (line), its annual echo readings (dots) and the VARC-3 threshold. Measurement noise creates false alarms; a label needs two consecutive abnormal echoes and arrives late.

Durability differs by valve

Severe SVD at 10 years: 10.0% after SAVR vs 1.5% after TAVI (NOTION); the SAVR-TAVR question is still open.

The label is noisy

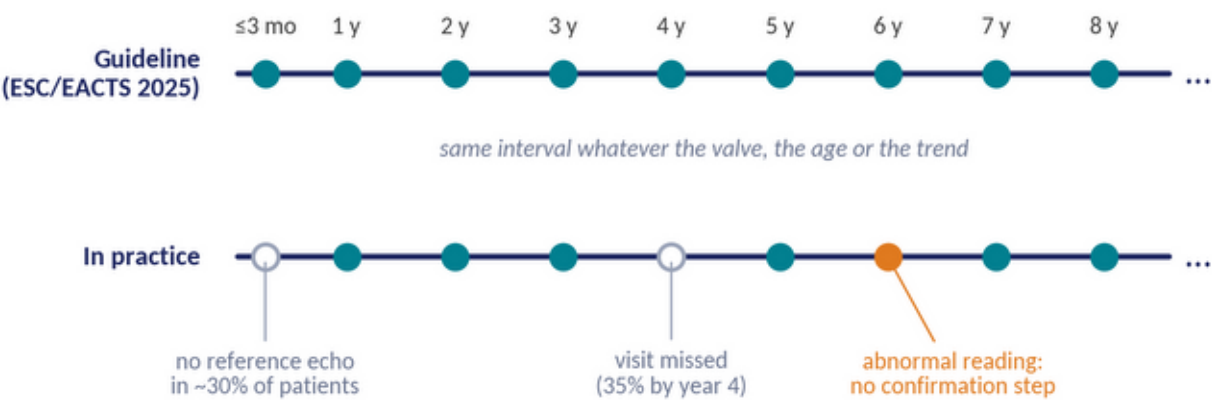
59–65% of first SVD calls are reversed at the next echo (Velders 2024).

Confirmation is late

Under annual surveillance, moderate deterioration is confirmed a mean 1.5 years after onset; 2.9 years for one patient in ten (our calibrated simulation).

Current workflow

One calendar for every patient



Guideline schedule: echo within 3 months, at 1 year, then every year (ESC/EACTS 2025); ACC/AHA similar.

Same echo for everyone

The interval does not depend on the valve model, the patient's age or the trend of the readings.

Timing errors are expensive

Urgent redo surgery: 22.6% mortality vs 1.4% when elective (Cartlidge 2019).

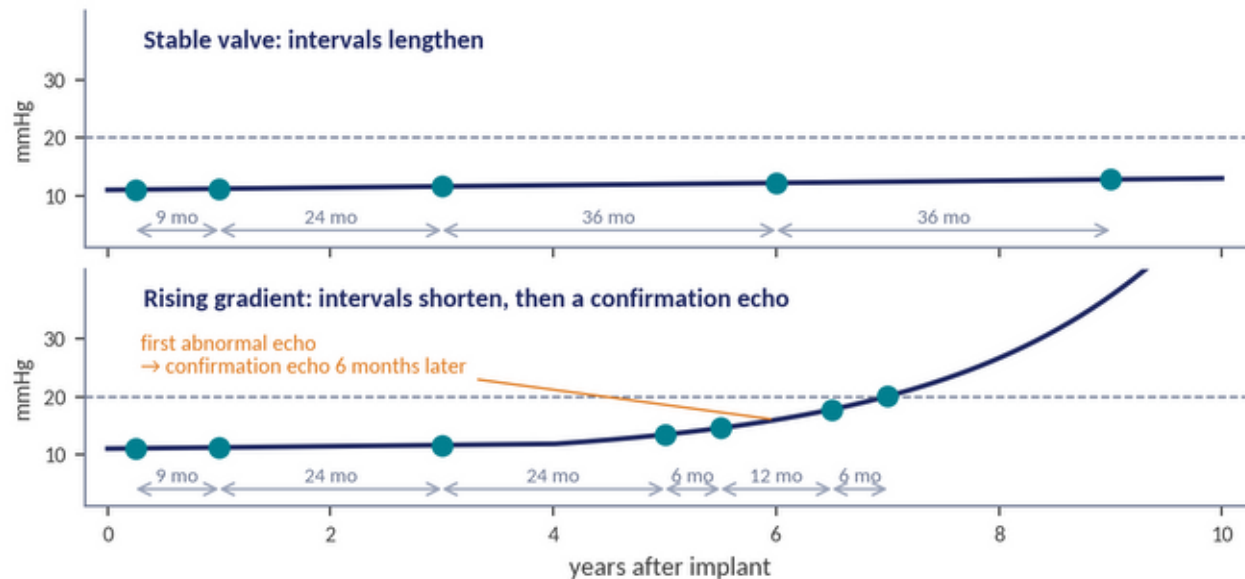
Effort in the wrong places

Low-risk patients are over-tested while valve-in-valve volume heads towards 42,000 a year by 2035 (Genereux 2025).

Hypothesis

Let the interval follow the risk

A risk estimate updated at every echo can lengthen the surveillance interval for stable valves and shorten it for valves that begin to deteriorate, without losing time to detection.



Schematic: two valves under a risk-adapted schedule. Dots are echoes; arrows show the chosen interval.

Targets, fixed before any analysis

$\geq 25\%$ fewer echoes per patient-decade

≤ 3 months added detection delay, on average and for the slowest-detected tenth

Earlier confirmation for the high-risk valves where timing matters most

Any single abnormal echo still triggers a confirmation echo; symptoms always override the schedule.

Study design

Who is in, what counts as deterioration, and what the truth is

Population

Adults with a first bioprosthetic SAVR or TAVR; valve model and size recorded; reference echo within 3 months (or a model $\times$ size norm, flagged); at least one follow-up echo.

Excluded: mechanical, homograft and Ross valves; index valve-in-valve; multiple prostheses.

$\geq 5,000$ valves from ≥ 3 centres; $\geq 2,000$ external.

Endpoints

Primary: VARC-3 moderate SVD (stage ≥ 2) confirmed on two consecutive echoes; death, endocarditis and non-SVD reintervention compete; incidence at 5, 8 and 10 years.

Secondary: valve failure; reintervention, elective vs urgent; gradient progression; echoes used and detection delay under each schedule.

Ground truth, highest level wins

1. Explant or autopsy pathology
2. Reintervention for SVD
3. VARC-3 criteria on two consecutive echoes
4. Blinded heart-team adjudication

Endocarditis, thrombosis, paravalvular leak and mismatch are adjudicated out.

Data

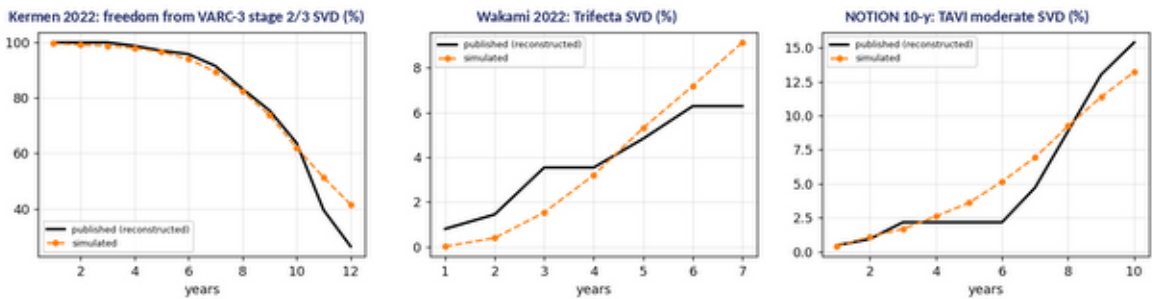
Registry-shaped for deployment; a calibrated simulation for the prototype

Deployment: routinely collected, linked by one key

Echo reporting database	serial gradients, EOA, DVI, LVEF
STS ACSD, STS/ACC TVT Registry	valve model, size, comorbidities
EHR notes, local NLP	atrial fibrillation, anticoagulation, symptoms
CMS claims, National Death Index	reintervention and death elsewhere

Needs exact dates and ages (a HIPAA limited data set). In the provided notes, a prosthetic echo with a gradient was found for only 36% of patients.

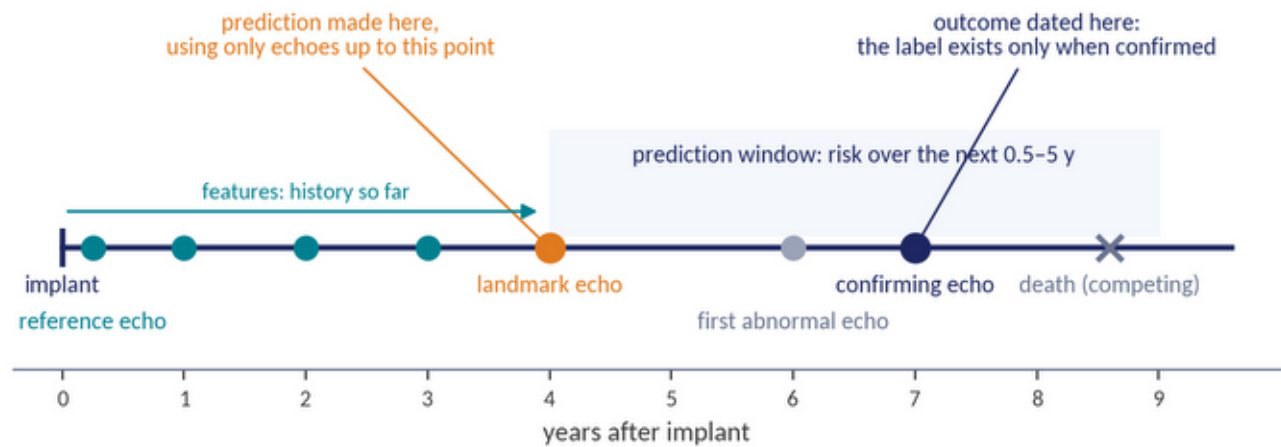
Prototype: no public serial-echo dataset, so we built one



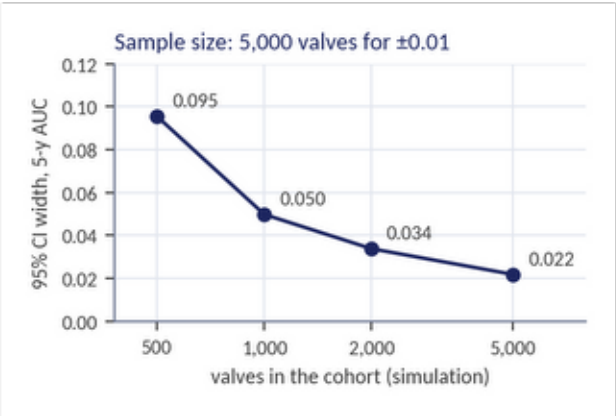
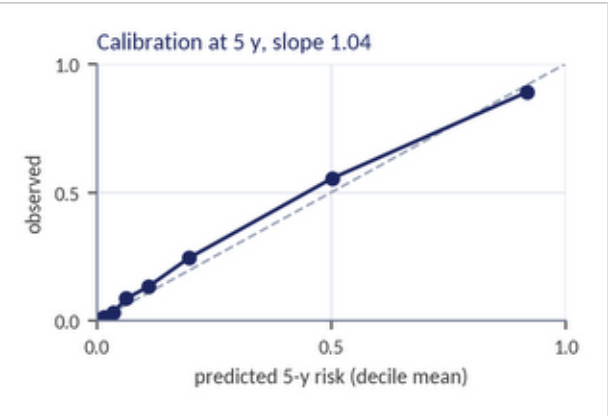
Simulated (orange) vs reconstructed published curves (black): within 1–3 points. NOTION's surgical arm predicted with no free parameter: 21.6% vs 20.7% at 10 y.

Validation

No peeking at the future, and death is a competing risk



Every follow-up echo is a landmark: features use only the history up to it; the outcome is dated at the confirming echo.



Design

Split by valve, 70 / 15 / 15, bootstrap by valve. Death, endocarditis and other reintervention as competing risks. Temporal and external validation in the real study.

Metrics

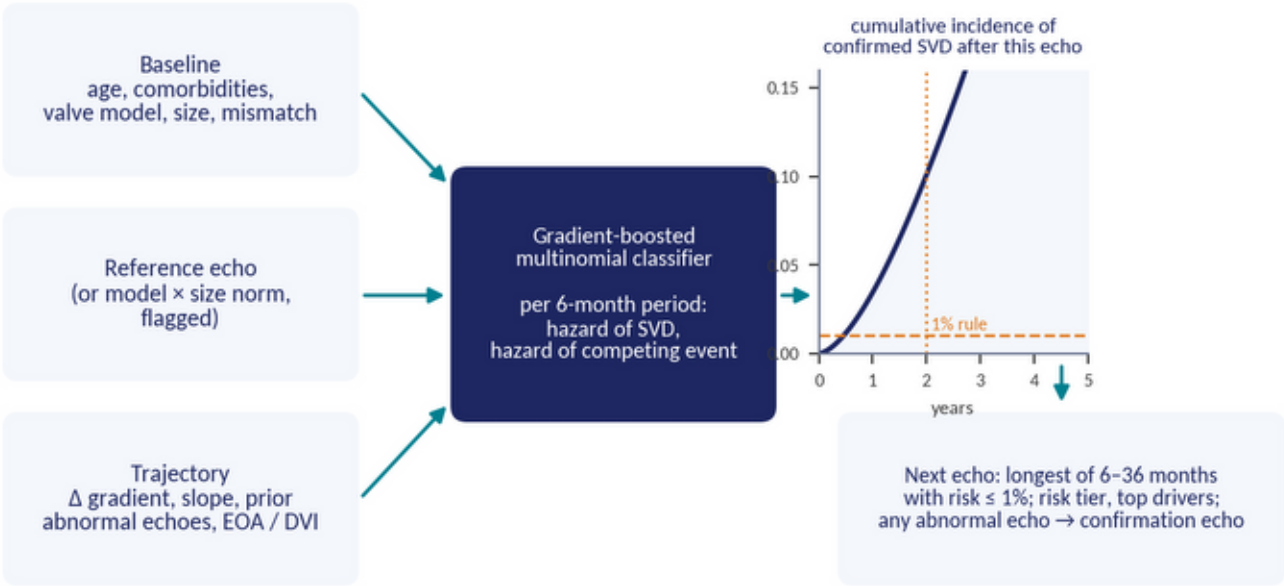
Time-dependent AUC and C-index; calibration slope; echoes per patient-decade and delay to confirmed detection. Reported separately on echoes not yet abnormal.

Comparators

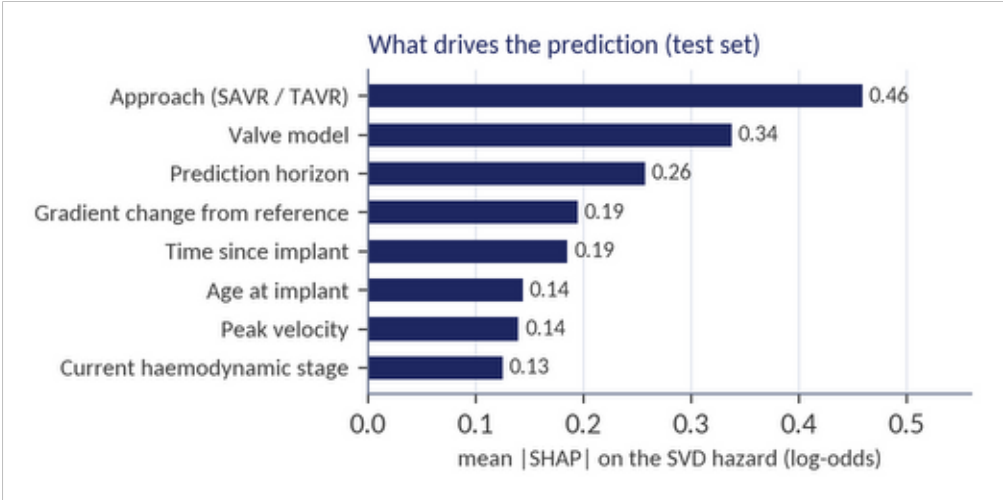
Time since implant alone; the published risk-factor Cox model; for surveillance, the guideline schedule with and without a confirmation echo.

Model

A hazard at every echo, turned into a date for the next one



Discrete-time cause-specific hazard model: survival × hazard summed over 6-month periods gives the cumulative incidence over any horizon, so one model answers "what is the risk if we wait 6, 12, 24 or 36 months?"

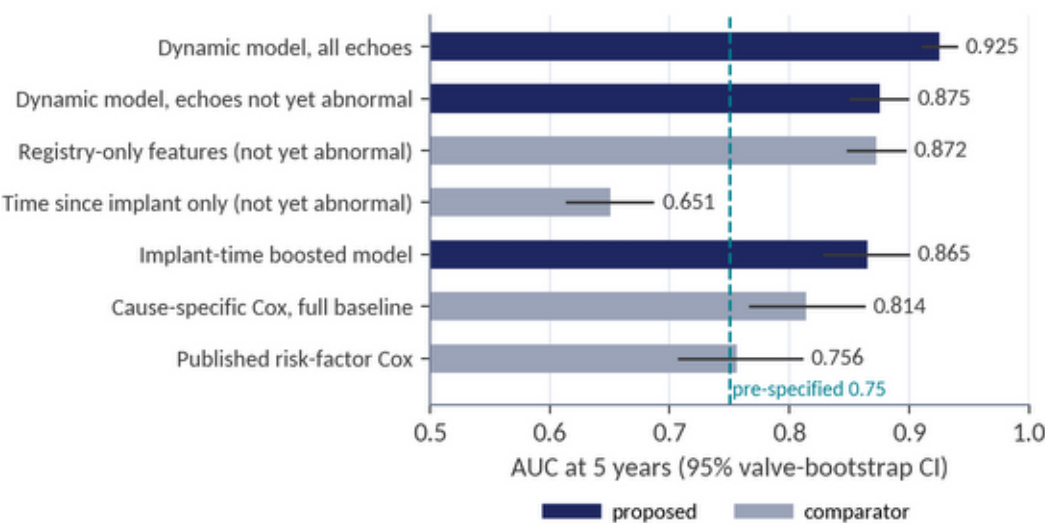


What the clinician sees

Risk at 1–5 years, a tier, the top drivers and the next echo interval. Test-set example: "9.9 y after TAVR: 5-y risk 5.1%, moderate; drivers TAVR ↓, gradient +4.9 mmHg ↑, time since implant ↑."

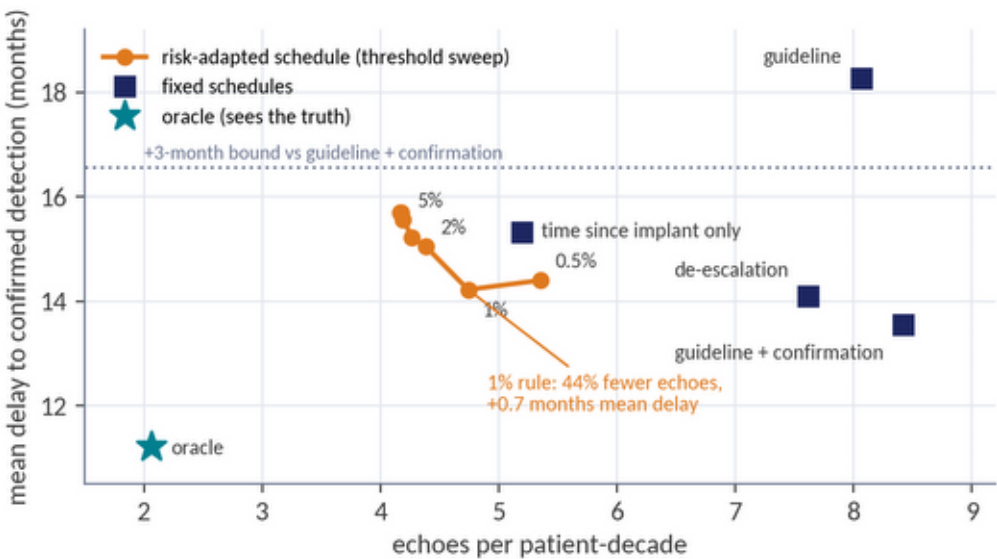
Results on the synthetic test set

1,401 held-out valves; every pre-specified bar cleared



Prediction

AUC 0.875 (0.85–0.90) on echoes not yet abnormal, against 0.65 for time since implant; calibration slope 1.04. At implant, 0.865 vs 0.756 for the published risk-factor Cox.



Surveillance

The confirmation echo alone cuts the mean delay from 1.5 to 1.1 years. The 1% rule then uses 44% fewer echoes than guideline + confirmation at +0.7 months mean delay, inside the 3-month bound.

Audit against the simulator's truth: even the confirmed VARC-3 label detects 60% of true deterioration, median lag 1.5 years. On 26 real de-identified patients the label rule was 86% sensitive and 89% specific (face validity only).

Impact and next steps

What changes, and what it will take to prove it

What it enables

A surveillance interval per valve, recomputed at every echo, with a confirmation rule built in; and an audit of the SVD definitions against the truth.

Next three months

Run the same pipeline on one institution's echo database (the schema is registry-shaped), then evaluate it in silent mode.

What validation needs

A linked echo database with STS/TVT at three or more centres, with exact dates and ages; then external validation in a second health system.

Readiness

Code and reports are public and reproducible. Real-world performance is unproven until external validation; the system schedules surveillance and never recommends reintervention.

Team CardioNTUA · github.com/abarmper/Dyania-Docathon · [report/main.pdf](#)